

GLACIER PARK, MT, USA

WMO: 727790

Lat: 48.304N

Lon: 114.264W

Elev: 901

StdP: 90.96

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24146

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.3	-16.7	-25.1	0.4	-19.2	-21.4	0.6	-15.9	11.3	-7.6	9.3	-7.3	1.7	10	0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	32.5	16.9	30.7	16.4	28.8	15.8	18.4	28.4	17.4	27.6	16.6	26.7	3.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.1	11.9	20.8	13.9	11.0	19.8	12.8	10.3	19.2	55.2	28.6	52.0	27.4	49.5	26.7	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	9.1	7.8	6.8	DB	-25.7	35.3	4.1	1.7	-28.7	36.5	-31.0	37.5	-33.3	38.4	-36.3	39.6
(5)				WB	-26.0	20.6	4.0	1.4	-28.8	21.6	-31.2	22.4	-33.5	23.2	-36.4	24.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.5	-4.2	-3.0	1.6	6.2	11.1	14.5	18.8	18.0	12.9	5.7	0.3	-4.2
	DBStd	9.20	5.83	5.42	4.42	3.51	3.66	3.49	3.14	3.10	3.74	3.90	4.61	5.41
	HDD10.0	2109	442	363	259	122	31	4	0	0	14	140	293	441
	HDD18.3	4401	700	596	518	363	225	123	32	43	168	391	542	699
	CDD10.0	844	0	0	0	9	64	139	274	249	100	8	1	0
	CDD18.3	93	0	0	0	0	1	8	47	34	3	0	0	0
	CDH23.3	2424	0	0	0	4	64	234	1072	869	179	1	0	0
	CDH26.7	900	0	0	0	0	9	65	440	341	45	0	0	0
Wind	WSAvg	2.2	1.8	2.1	2.6	3.0	2.9	2.5	2.3	2.3	2.1	1.9	1.9	1.7
Precipitation	PrecAvg	437	39	27	27	30	50	63	31	33	33	26	35	42
	PrecMax	641	87	50	75	66	121	144	153	106	101	64	83	119
	PrecMin	267	5	1	2	7	6	12	1	0	0	1	7	13
	PrecStd	91	18	12	13	14	27	34	28	27	22	16	20	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	9.8	16.4	23.4	28.1	31.7	34.9	34.0	30.7	21.7	13.6	8.3
		MCWB	3.9	5.4	7.9	11.5	14.5	17.2	17.8	16.9	15.7	12.4	9.3	5.3
	2%	DB	5.0	6.5	12.4	19.9	25.1	28.4	33.0	32.1	27.7	18.3	10.7	5.5
		MCWB	2.7	3.3	6.0	10.2	13.6	15.8	17.3	16.4	14.7	10.7	6.8	3.3
	5%	DB	3.7	4.8	10.2	16.7	22.7	26.0	31.2	30.5	25.3	15.3	8.6	3.8
		MCWB	1.8	2.2	5.1	8.5	12.5	15.0	16.9	16.0	13.9	9.1	5.7	1.9
	10%	DB	2.7	3.5	8.3	13.8	20.0	23.7	29.2	28.5	22.6	12.8	6.9	2.4
		MCWB	1.1	1.3	4.0	7.1	11.3	14.0	16.3	15.5	13.0	8.0	4.4	1.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.4	5.7	8.6	12.3	15.7	18.5	20.3	19.0	16.9	13.5	10.4	5.5
		MCDB	6.2	8.4	14.2	21.1	24.8	28.9	30.1	28.7	26.0	20.1	12.9	7.7
	2%	WB	3.0	3.8	7.0	10.6	14.4	16.9	18.9	17.6	15.6	11.4	7.5	3.5
		MCDB	4.5	6.0	11.5	18.4	23.1	26.3	29.0	28.3	25.3	16.6	10.1	5.1
	5%	WB	2.1	2.5	5.6	9.1	13.2	15.8	17.9	16.8	14.6	10.0	6.1	2.2
		MCDB	3.4	4.2	9.3	15.6	21.0	24.1	28.0	27.5	23.8	14.2	8.3	3.5
	10%	WB	1.3	1.6	4.4	7.7	11.9	14.7	17.0	16.1	13.4	8.7	4.8	1.2
		MCDB	2.4	3.2	7.7	13.3	18.6	22.0	27.0	26.5	21.6	12.3	6.7	2.1
Mean Daily Temperature Range	5% DB	MDBR	7.2	8.7	10.3	12.4	13.7	14.0	18.2	18.6	16.3	12.6	8.1	6.5
		MCDBR	7.4	9.5	14.2	18.6	19.5	19.6	22.0	22.2	21.7	17.3	10.8	7.0
		MCWBR	5.7	6.7	8.7	9.9	9.2	8.7	8.5	8.7	10.0	10.3	7.6	5.5
	5% WB	MCDBR	6.8	8.7	12.6	16.8	17.4	17.5	18.8	19.2	19.9	14.9	9.6	6.6
		MCWBR	5.3	6.4	8.3	9.3	8.5	8.1	8.0	7.9	9.5	9.1	7.1	5.5
Clear-Sky Solar Irradiance	taub	0.251	0.260	0.280	0.319	0.325	0.329	0.333	0.344	0.318	0.293	0.275	0.251	
	taud	2.395	2.374	2.416	2.408	2.493	2.523	2.518	2.482	2.530	2.594	2.530	2.420	
	Ebn at Noon	842	914	947	934	937	931	922	897	894	870	811	795	
	Edh at Noon	65	84	95	107	103	101	100	99	85	68	58	56	
All-Sky Solar Radiation	RadAvg	1.09	2.02	3.13	4.48	5.45	5.95	7.07	5.83	4.08	2.41	1.20	0.85	
	RadStd	0.12	0.20	0.34	0.39	0.43	0.56	0.52	0.40	0.42	0.27	0.11	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+64	N/A
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page